

Wk 8 Schottky Diodes

Chapter 5:

Schottky Diodes

Streetman and Banerjee, 'Solid State Electronic Devices,' 6th ed., Pearson, 2006.

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Questions?

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Metal-Semiconductor Junctions

CO 7: Basic understanding of Schottky diodes and thermionic emission.

1. Schottky diode

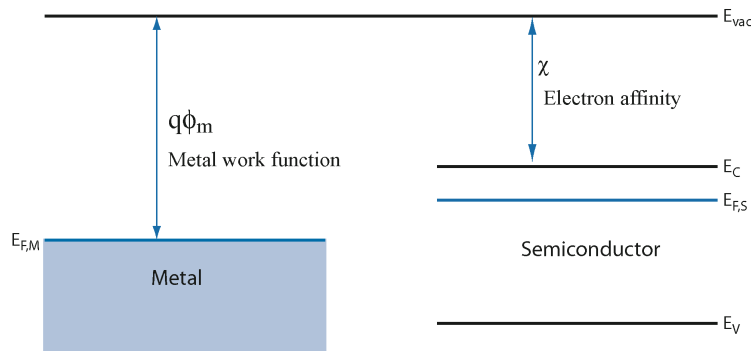
1. Metal workfunction
2. Electron affinity
3. Schottky barrier height
4. Band alignments
5. Electrostatics (Depletion width, capacitance)
6. Current flow

2. Ohmic Contact

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Metal-Semiconductor Junction (Schottky Diode)

- Energies and alignments from the vacuum level:

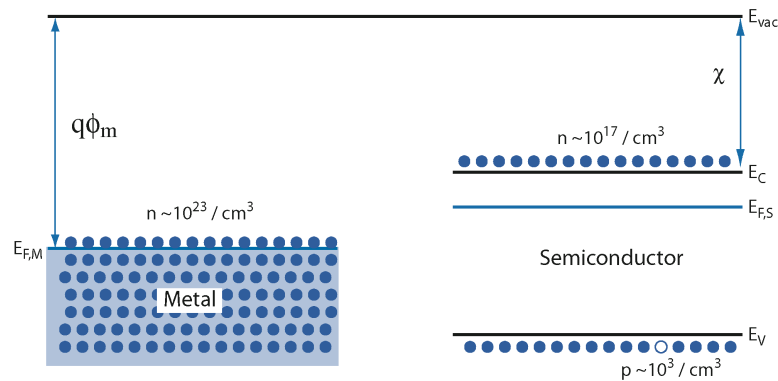


- Metal workfunction Φ_m : distance from the vacuum level to the metal Fermi level.
- Electron affinity χ : distance from the semiconductor conduction band edge to the vacuum. This value is a property of the material (Si), and it is, therefore, the same at any point in the band diagram.
- The vacuum level, E_{vac} , is a continuous function of position (no discrete jumps, no step functions).

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Metal-Semiconductor Junction (Schottky Diode)

- Electron and hole concentrations.

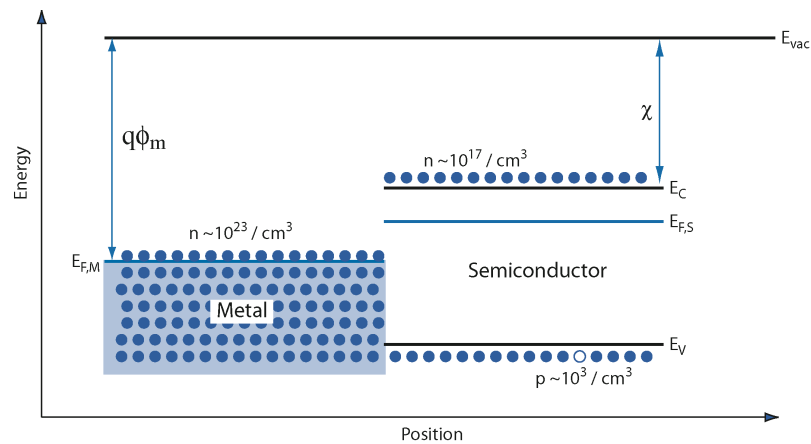


- The electron concentration in the metal is $\sim 10^{23} / \text{cm}^3$.
- The electron concentration in the semiconductor is $\sim 10^{17} / \text{cm}^3$.
- The hole concentration in the semiconductor is $\sim 10^3 / \text{cm}^3$.

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Metal-Semiconductor Junction Potential

- Band diagram at $t=0$, before charge transfer occurs.

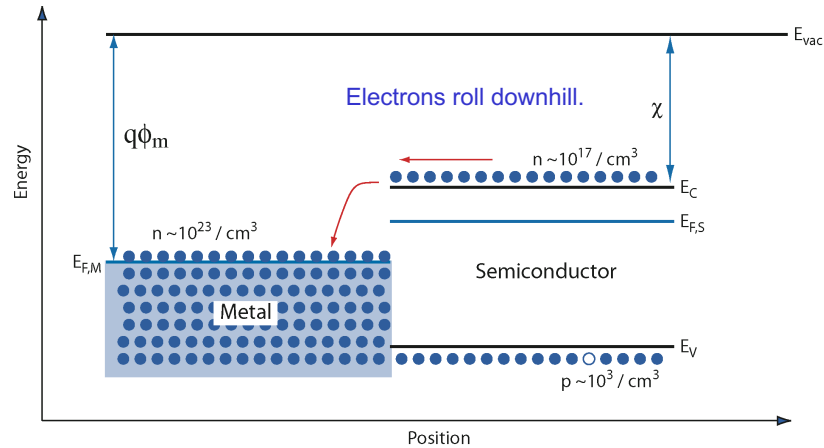


- Now, bring the metal and semiconductor into intimate contact so that they can exchange electrons and holes. The above diagram is at $t=0$, the moment they come into contact, but before any charge transfer takes place.
- What happens next?

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Metal-Semiconductor Junction Potential

- Charge transfer

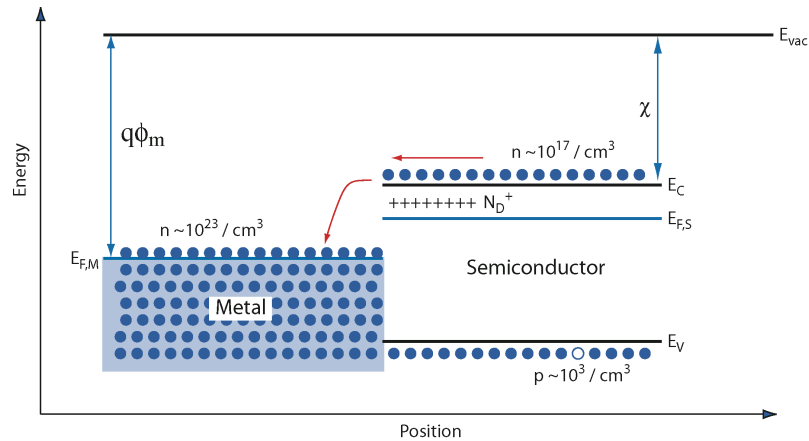


- What causes the flow to stop?

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Metal-Semiconductor Junction (Schottky Diode)

- Junction potential

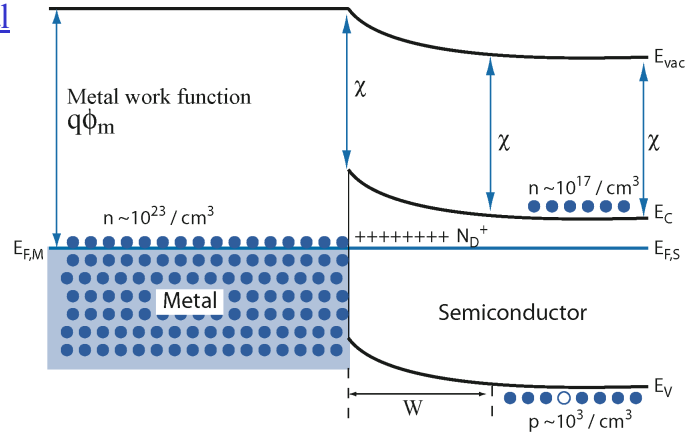


- Create a depletion region near the metal-semiconductor junction with positive fixed charge density N_D^+ .
- So what?

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Metal-Semiconductor Junction (Schottky Diode)

• Junction potential



- An electric field / built-in potential develops at the junction stopping the flow of electrons from the semiconductor to the metal.
- The Fermi levels are now aligned as they must be in equilibrium.
- Looks like a one-sided p⁺n junction with all of the band bending in the n-side of the junction.

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Metal-Semiconductor Junction Potential

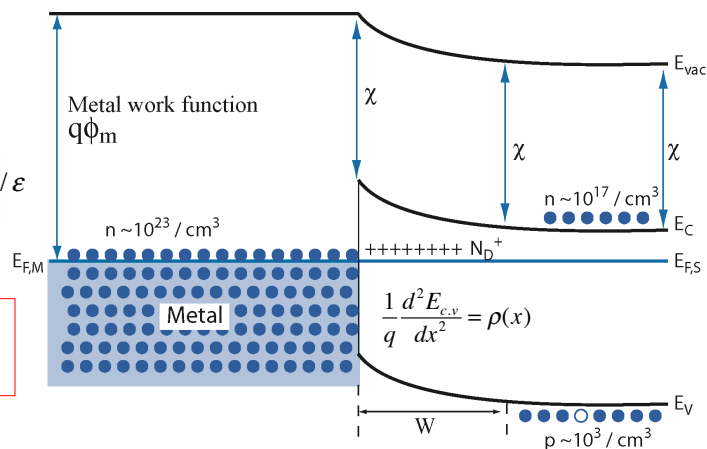
• Band curvature

$$\nabla \cdot \mathcal{E} = \rho(\mathbf{r}) / \epsilon$$

$$\nabla \cdot (-\nabla V) = \rho(\mathbf{r}) / \epsilon$$

$$-\frac{d^2 V}{dx^2} = \frac{1}{q} \frac{d^2 E_c}{dx^2} = \rho(x) / \epsilon$$

The sign of the second derivative is the sign of the charge.

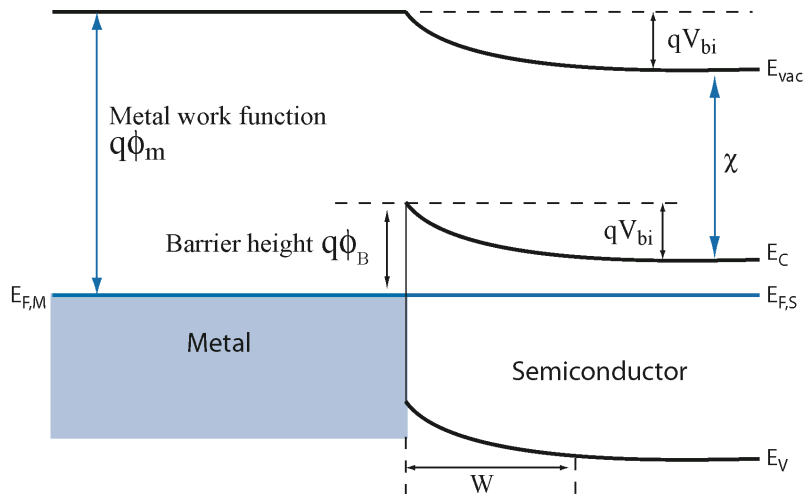


- In a region of positive charge, the conduction and valence band are 'concave up.' The second derivative of the conduction band or valence band with respect to x is positive.
- In a region of negative charge, the bands are 'concave down.' The second derivatives are negative.
- If there is no curvature (if the bands are linear), there is no net charge.

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Metal-Semiconductor Junction (Schottky Diode)

- Metal-semiconductor barrier height (Schottky barrier height), ϕ_B
- Built-in voltage, V_{bi}



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Metal Workfunctions

- Property of the metal

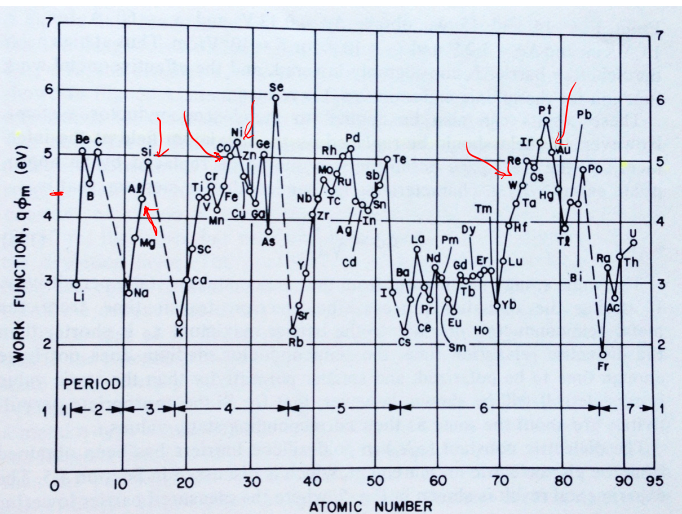


Fig. 4 Metal work function for a clean metal surface in a vacuum versus atomic number. Note the periodic nature of the increase and decrease of the work functions within each group. (After Michaelson, Ref. 9.)

From S. Sze, 'Physics of Semiconductors,' 2nd ed., John Wiley & Sons, 1981, p. 251.

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Metal-Semiconductor Junction (Schottky Diode)

- Schottky Barrier height Φ_B :

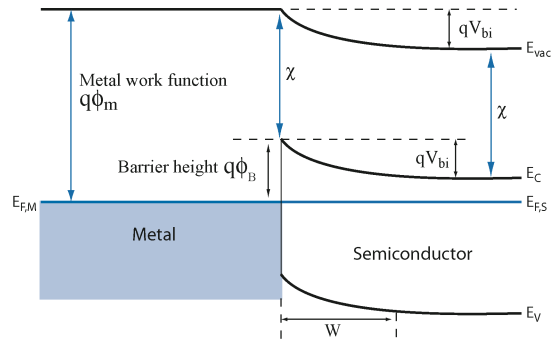
$$q\phi_B = q\phi_m - \chi$$

- Built-in voltage V_{bi} :

$$qV_{bi} = q\phi_B - (E_c - E_F)_{\text{bulk}} = q\phi_m - \chi - (E_c - E_F)_{\text{bulk}}$$

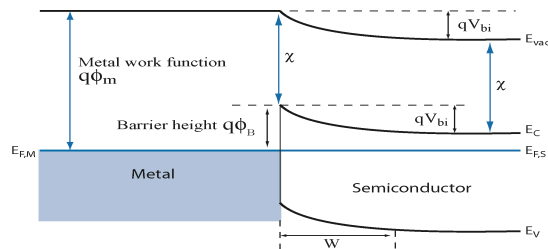
$$n = N_D = N_C e^{-(E_c - E_F)/kT} \Rightarrow -(E_c - E_F) = kT \ln\left(\frac{N_D}{N_C}\right)$$

$$qV_{bi} = q\phi_m - \chi + kT \ln\left(\frac{N_D}{N_C}\right)$$



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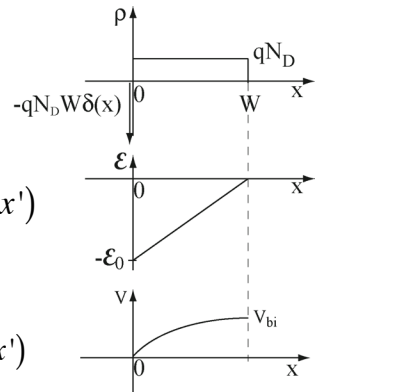
Metal-Semiconductor Junction Electrostatics



Screening length in the metal is on the order of a few Å.

$$\mathcal{E}(x) = \frac{1}{\epsilon_s} \int_0^x dx' \rho(x')$$

$$V(x) = -\int_0^x dx' \mathcal{E}(x')$$



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Metal-Semiconductor Junction Electrostatics

- Depletion Width, W
- Capacitance, ϵ_s / W .
- Just like one-sided pn junction.

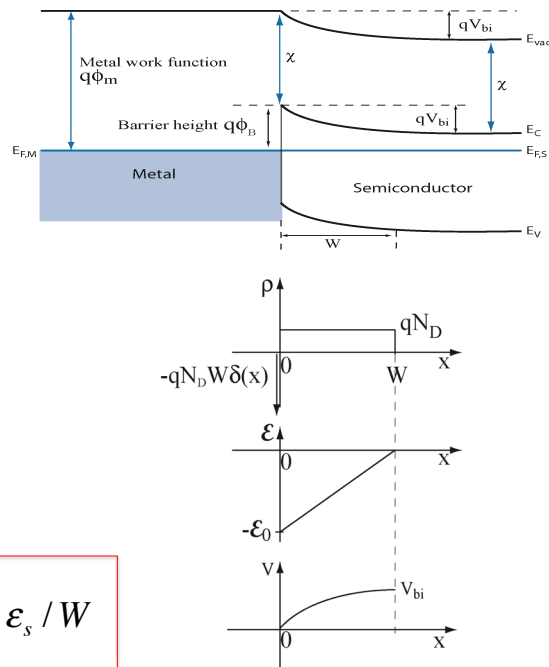
$$qV_{bi} = q\phi_m - \chi + kT \ln\left(\frac{N_D}{N_C}\right)$$

$$\epsilon_0 = \frac{qN_D W}{\epsilon_s}$$

$$V_{bi} = \frac{1}{2} \epsilon_0 W = \frac{qN_D W^2}{2\epsilon_s}$$

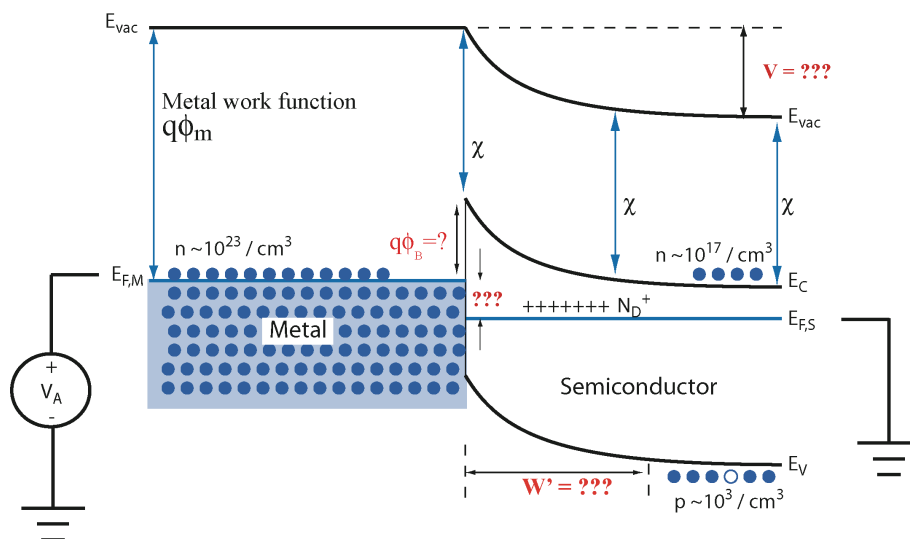
$$W = \sqrt{\frac{2\epsilon_s V_{bi}}{qN_D}}$$

$$C_j = \epsilon_s / W$$



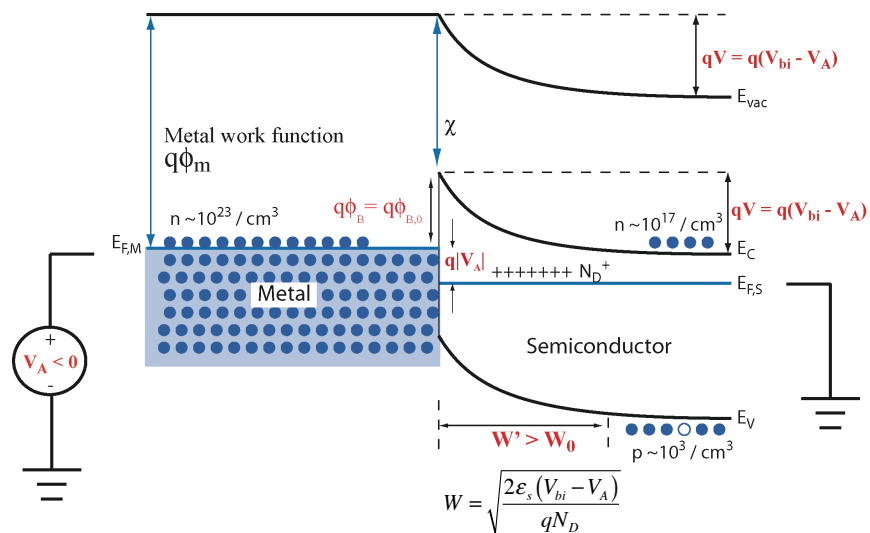
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Metal-Semiconductor Junction Under Bias



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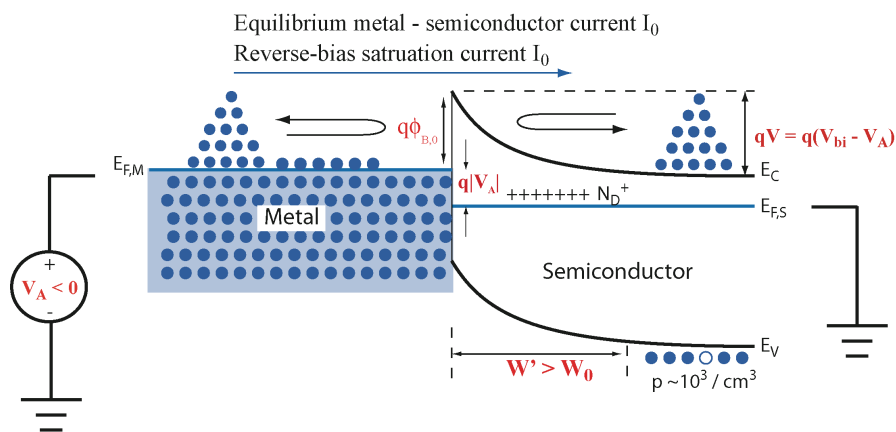
Schottky Diode Under **REVERSE** Bias



- The metal and semiconductor quasi-Fermi levels split at the metal-semiconductor junction by the magnitude of the applied bias, $q|V_A|$.
- The **Schottky barrier height is unchanged**.
- The voltage drop across the semiconductor increases to $V_{bi} - V_A$, just like in a pn junction.
- The depletion width W increases.

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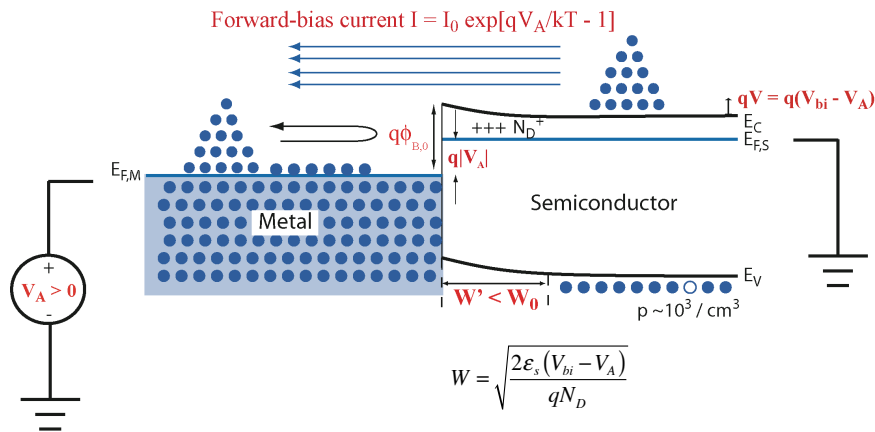
Schottky Diode **Current** Under **REVERSE** Bias



- The **Schottky barrier height is unchanged**. Thus the equilibrium injection rate from the metal to the semiconductor remains the same. The electrons in the metal 'see' the same potential barrier as in equilibrium.
- The barrier for the electrons in the semiconductor increases to $V_{bi} + |V_A|$. Thus, the flow of electrons from the semiconductor to the metal is reduced.
- For $|V_A| > \text{a few } kT/q$, the ideal reverse bias current is independent of $|V_A|$ and equal to I_0 just like in the ideal pn diode.

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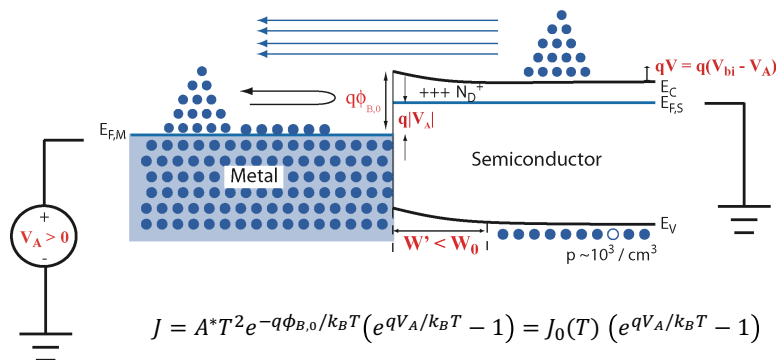
Schottky Diode Current Under FORWARD Bias



- Forward bias reduces the barrier height $q(V_{bi} - V_A)$ for electrons to flow from the semiconductor to the metal.
- Again, the barrier height for electron flow from the metal to the semiconductor is unchanged $q\phi_{B,0}$.
- The DC current equation is the same as the pn diode: $I = I_0 \exp[qV_A/kT - 1]$
- The current flow is thermionic emission 'over the barrier' flow.
- The current is carried by MAJORITY CARRIERS.

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Schottky Diode Current – Thermionic Emission



$$J = A^* T^2 e^{-q\phi_{B,0}/k_B T} (e^{qV_A/k_B T} - 1) = J_0(T) (e^{qV_A/k_B T} - 1)$$

$$A^* = \frac{4\pi q m^* k_B^2}{h^3} \left(\frac{\text{A}}{\text{cm}^2 \text{K}^2} \right) \quad \text{Richardson's constant}$$

$$A^* \frac{m_0}{m^*} = 120 \left(\frac{\text{A}}{\text{cm}^2 \text{K}^2} \right)$$

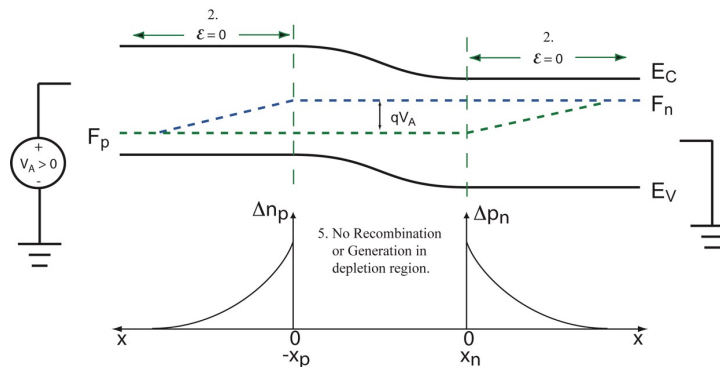
S. M. Sze, Sec. 5.4.1

$$\text{For transport along } z, m^* = \sqrt{m_x^* m_y^*}$$

$$\text{For } V_A > 3k_B T: J = A^* T^2 \exp\left(-\frac{q(\phi_{B,0} - V_A)}{k_B T}\right)$$

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pn-diode vs. Schottky diode switching time



$$I = \frac{q}{\tau_p} \int_0^\infty dx \Delta p_n(x) = \frac{q}{\tau_p} \int_0^\infty dx \Delta p_n(0) e^{-x/L_p} = \frac{q \Delta p_n(0) L_p}{\tau_p} = q n_{p0} \frac{L_p}{\tau_p} (e^{qV_A/KT} - 1)$$

$$J = q \left[n_{p0} \frac{L_n}{\tau_n} + p_{n0} \frac{L_p}{\tau_p} \right] (e^{qV_A/KT} - 1)$$

- The current of a pn diode is limited by the decay time of the minority carriers. A relatively slow time $\sim 1 \mu s$.
- The switching response is also limited by that decay time.

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Schottky diode vs. pn-junction diode

- **Schottky** diodes are MAJORITY carrier devices and the response of majority carriers is extremely fast.

$$\tau = \frac{\epsilon}{\sigma} \approx 10 \text{ fs}$$

- **pn diode** response is governed by the **MINORITY** carrier lifetime which is much slower.

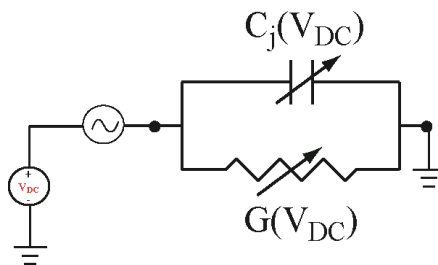
$$\tau \approx 1 \mu s$$

- Same DC response, but Schottky diodes can operate at much higher frequencies. The highest of all electronic devices.
- Schottky diodes are used for ultra-high frequency applications. THz detectors.

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Low-Frequency, AC Model of Diode

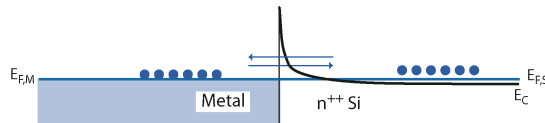
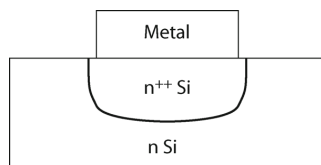
- Small signal, low frequency model
- Conductance: $G_0 \equiv \frac{dI}{dV} = I_0 \frac{d}{dV} e^{qV/kT} \underset{V \geq 3kT/q}{=} \frac{I}{kT/q} = \frac{I_0}{0.026 \text{ V}} e^{qV/kT}$
- For $V > 3kT/q$, $G = I/(kT/q) = I / (26 \text{ mV})$.
- $C = C_j = \epsilon_s/W$.



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Schottky contact vs. Ohmic Contact

- How are they different?
- During fabrication, the ohmic contact is almost always annealed.
- The semiconductor of an ohmic contact is ALWAYS heavily doped as possible.

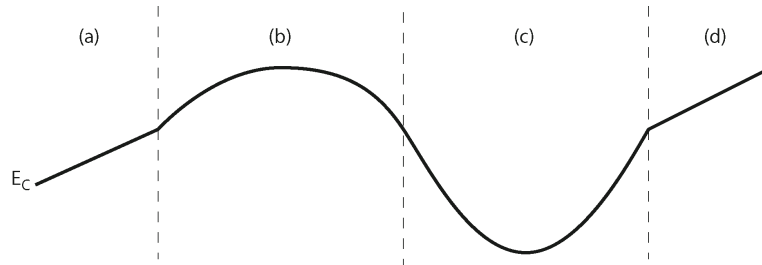


- The depletion width W becomes so narrow, that electrons can tunnel through the barrier. The barrier becomes so narrow that it is no longer a barrier.

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Retrieval

- Given the conduction band below, in each region (a) – (d), determine whether the net charge is positive, negative, or zero.

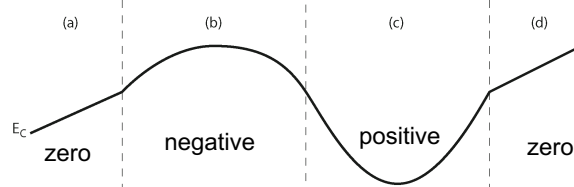


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Retrieval

- Given the conduction band below, in each region (a) – (d), determine whether the net charge is positive, negative, or zero.

$$\frac{d^2 E_c}{dx^2} \propto \rho(x)$$



- In a region of positive charge, the conduction and valence band are 'concave up.' The second derivative of the conduction band or valence band with respect to x is positive.
- In a region of negative charge, the bands are 'concave down.' The second derivatives are negative.
- If there is no curvature (if the bands are linear), there is no charge.

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Retrieval

- Your company is having trouble making an ohmic contact to a layer of n-type Si doped $10^{16} / \text{cm}^3$. What would you suggest?

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Retrieval

- You need a diode that can operate at frequencies of several hundred GHz. What do you recommend?

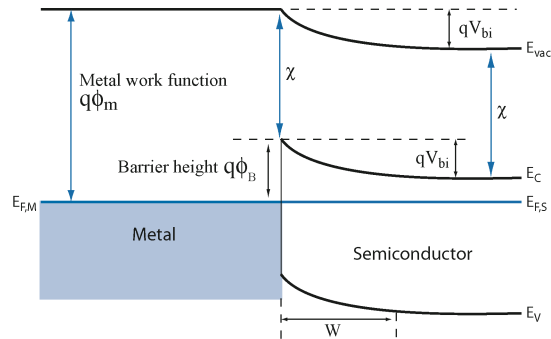
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Schottky Diode Summary

- Schottky Barrier height Φ_B :

$$q\phi_B = q\phi_m - \chi$$

$$qV_{bi} = q\phi_m - \chi + kT \ln\left(\frac{N_D}{N_C}\right)$$



$$V_{bi} = \frac{1}{2} \epsilon_0 W = \frac{qN_D W^2}{2\epsilon_s}$$

$$W = \sqrt{\frac{2\epsilon_s V_{bi}}{qN_D}}$$

$$C_j = \epsilon_s / W$$

$$J = A^* T^2 \exp\left(-\frac{q\phi_B}{k_B T}\right) \left[\exp\left(\frac{qV}{k_B T}\right) - 1\right] = J_0(T) \left[\exp\left(\frac{qV}{k_B T}\right) - 1\right]$$

$$A^* = \frac{4\pi q m^* k_B^2}{h^3} \left(\frac{A}{\text{cm}^2 \text{K}^2}\right) \quad \text{Richardson's constant}$$

$$A^* \frac{m_0}{m^*} = 120 \left(\frac{A}{\text{cm}^2 \text{K}^2}\right)$$

$$J = A^* T^2 \exp\left(-\frac{q(\phi_B - V)}{k_B T}\right) \quad (3k_B T/q < V < \phi_B)$$